

**Creating a table pre-filled with
the letters of the alphabet.**

A B C D E F G H I J K

01 LetterTable.

02 TableValues.

03 FILLER PIC X(13) VALUE "ABCDEF~~GH~~IJKLM".

03 FILLER PIC X(13) VALUE "NOPQR~~ST~~UVWXYZ".

First, we reserve an area of storage and use the VALUE clause to fill it with the letters of the alphabet.

Letter(1) Letter(2) Letter(3) Letter(4) Letter(5) Letter(6) Letter(7) Letter(8) Letter(9) Letter(10) Letter(11)

A	B	C	D	E	F	G	H	I	J	K
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01 LetterTable.
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02 TableValues.
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03 FILLER PIC X(13) VALUE "ABCDEFGHJKLM".
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03 FILLER PIC X(13) VALUE "NOPQRSTUVWXYZ".
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02 FILLER REDEFINES TableValues.
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03 Letter PIC X OCCURS 26 TIMES.
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Then, we use the REDEFINES clause to redefine the area of memory as a table. In effect we impose a table description on the area of storage.

**Creating a table pre-filled with
the months of the year.**

January February March April



01 MonthTable.

02 MonthValues.

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03 FILLER      PIC X(18) VALUE "January February".
03 FILLER      PIC X(18) VALUE "March April".
03 FILLER      PIC X(18) VALUE "May June".
03 FILLER      PIC X(18) VALUE "July August".
03 FILLER      PIC X(18) VALUE "SeptemberOctober".
03 FILLER      PIC X(18) VALUE "November December".
```

First we reserve an area of storage and use the VALUE clause to fill it with the names of the months.

Month(1)

Month(2)

Month(3)

Month(4)

January

February

March

April



01 MonthTable.

02 MonthValues.

03 FILLER PIC X(18) VALUE "January February".

03 FILLER PIC X(18) VALUE "March April".

03 FILLER PIC X(18) VALUE "May June".

03 FILLER PIC X(18) VALUE "July August".

03 FILLER PIC X(18) VALUE "SeptemberOctober".

03 FILLER PIC X(18) VALUE "November December".

02 FILLER REDEFINES MonthValues.

03 Month OCCURS 12 TIMES PIC X(9) .

Then, we use the REDEFINES clause to redefine the area of memory as a table.

Notice that we use the word FILLER whenever we need to declare an item but don't need to refer to it by name in our program

Creating a two dimension table table pre-filled with the surcharge values for late videos.

Scenario

A video rental shop charges its customers a surcharge on late videos. The per-diem surcharge depends on how recently the video was released (Old, Recent, New) and the number of days the video is overdue (1-3, 4-7, >7). The surcharge values are held in a two dimension table.

The actual surcharge is calculated as -
 $\text{PerDiemSurcharge}(\text{VideoAge}, \text{OverduePeriod}) * \text{NumOfDaysOverdue}.$

050 100 150 075 125 200 100 150 300

01 LateVideoSurchargeTable.

02 SurchargeValues.

03 FILLER PIC X(6) VALUE "050100150".

03 FILLER PIC X(6) VALUE "075125200".

03 FILLER PIC X(6) VALUE "100150300".

First we reserve an area of storage and use the VALUE clause to fill it with the surcharge values.

1	2	3
050 100 150	075 125 200	100 150 300

01 LateVideoSurchargeTable.

02 SurchargeValues.

03 FILLER PIC X(6) VALUE "050100150".

03 FILLER PIC X(6) VALUE "075125200".

03 FILLER PIC X(6) VALUE "100150300".

02 FILLER REDEFINES SurchargeValues.

03 VideoAge OCCURS 3 TIMES.

Then, we use the first
OCCURS clause to lay down
the first dimension of the
table.

1			2			3		
1	2	3	1	2	3	1	2	1
050	100	150	075	125	200	100	150	300

```

01 LateVideoSurchargeTable.
  02 SurchargeValues.
    03 FILLER PIC X(6) VALUE "050100150".
    03 FILLER PIC X(6) VALUE "075125200".
    03 FILLER PIC X(6) VALUE "100150300".
  02 FILLER REDEFINES SurchargeValues.
    03 VideoAge OCCURS 3 TIMES.
      04 OverduePeriod OCCURS 3 TIMES.
        05 PerDiemSurcharge PIC 9V99.

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Pay particular attention to the way the two dimension table maps on to the pre-filled area of storage